Chapter 8

Problems & Solutions

1)

P.8-1 Consider lossless stripline designs for a given characteristic impedance.

a) How should the dielectric thickness, d, be changed for a given plate width, w, if the dielectric constant, ϵ_r, is doubled?

b) How should w be changed for a given d if ϵ_r is doubled?

c) How should w be changed for a given ϵ_r if d is doubled?

d) Will the velocity of propagation remain the same as that for the original line after the changes specified in parts (a), (b), and (c)? Explain.

P. 8-1 Substituting Eqs. (8-17) and (8-18) in Eq. (8-43):

$$Z_0 = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}}$$

a) $Z_0 = \frac{d'}{w}\sqrt{\frac{\mu}{2\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow d' = \sqrt{2}\, d.$

b) $Z_0 = \frac{d}{w'}\sqrt{\frac{\mu}{2\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow w' = \frac{1}{\sqrt{2}} w.$

c) $Z_0 = \frac{2d}{w'}\sqrt{\frac{\mu}{\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow w' = 2w.$

d) $u_p = \frac{1}{\sqrt{\mu\epsilon}} \longrightarrow u_{pa} = u_p/\sqrt{2}$ for part a.

$u_{pb} = u_p/\sqrt{2}$ for part b.

$u_{pc} = u_p$ for part c.

P.8-2 Consider a transmission line made of two parallel brass strips—$\sigma_c = 1.6 \times 10^7$ (S/m)—of width 20 (mm) and separated by a lossy dielectric slab—$\mu = \mu_0$, $\epsilon_r = 3$, $\sigma = 10^{-3}$ (S/m)—of thickness 2.5 (mm). The operating frequency is 500 (MHz).

a) Calculate the R, L, G, and C per unit length.

b) Compare the magnitudes of the axial and transverse components of the electric field.

c) Find γ and Z_0.

P. 8-2 Given: $\sigma_c = 1.6\times10^{7}$ (S/m), $w = 0.02$ (m), $d = 2.5\times10^{-3}$ (m)
Lossy dielectric slab: $\mu=\mu_0$, $\epsilon_r=3$, $\sigma=10^{-3}$ (S/m).
$f = 5\times10^{8}$ (Hz).

a) $R = \frac{2}{w}\sqrt{\frac{\pi f\mu_0}{\sigma_c}} = 1.11$ (Ω/m).

$L = \mu\frac{d}{w} = 0.157$ $(\mu H/m)$.

$G = \sigma\frac{w}{d} = 0.008$ (S/m).

$C = \epsilon\frac{w}{d} = 0.212$ (nF/m).

b) $\frac{|E_z|}{|E_y|} = \sqrt{\frac{\omega\epsilon}{\sigma_c}} = 4.167\times10^{-5}$.

c) $\omega L = 493.5 \gg R$, $\omega C = 0.667 \gg G$.

$\gamma \cong j\omega\sqrt{LC}\left[1+\frac{1}{2j}\left(\frac{R}{\omega L}+\frac{G}{\omega C}\right)\right] = 0.129 + j18.14$ (m^{-1}),

$Z_0 \cong \sqrt{\frac{L}{C}}\left[1+\frac{1}{2j}\left(\frac{R}{\omega L}-\frac{G}{\omega C}\right)\right] = 27.21 + j0.13$ (Ω).

P.8-5 The following characteristics have been measured on a lossy transmission line at 100(MHz):

$Z_0 = 50 + j0$ (Ω),

$\alpha = 0.01$ (dB/m),

$\beta = 0.8\pi$ (rad/m).

Determine R, L, G, and C for the line.

P. 8-5 Eq. (8-38): $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$.

Given $Z_0 = 50 + j0$ (Ω) — Purely real.

$\mathrm{Im}(Z_0) = 0 \longrightarrow \frac{R}{L} = \frac{G}{C} = k$ (Distortionless line).

Given: $\alpha = 0.01$ (dB/m) $= 0.00115$ (Np/m).
$\beta = 0.8\pi$ (rad/m); $f = 10^{8}$ (Hz).

From Eqs. (8-48), (8-49) and (8-51): $\alpha = R\sqrt{\frac{C}{L}}$, $\beta = \omega\sqrt{LC}$, $Z_0 = \sqrt{\frac{L}{C}}$.

$R = \alpha Z_0 = 0.0576$ (Ω/m), $L = \frac{\beta Z_0}{2\pi f} = 0.20$ $(\mu H/m)$,

$G = \frac{RC}{L} = \frac{\alpha}{Z_0} = 23$ $(\mu S/m)$, $C = \frac{L}{Z_0^2} = 80$ (pF/m).

P.8-7 Express $V(z)$ and $I(z)$ in terms of the voltage V_i and current I_i at the input end and γ and Z_0 of a transmission line (a) in exponential form, and (b) in hyperbolic form.

P. 8-7 $V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$,

$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$.

At $z=0$; $V(0) = V_i = V_0^+ + V_0^-$, $I(0) = I_i = I_0^+ + I_0^- = \frac{1}{Z_0}(V_0^+ - V_0^-)$.

$\longrightarrow V_0^+ = \frac{1}{2}(V_i + I_i Z_0)$, $V_0^- = \frac{1}{2}(V_i - I_i Z_0)$.

a) $V(z) = \frac{1}{2}(V_i + I_i Z_0) e^{-\gamma z} + \frac{1}{2}(V_i - I_i Z_0) e^{\gamma z}$,

$I(z) = \frac{1}{2Z_0}(V_i + I_i Z_0) e^{-\gamma z} - \frac{1}{2Z_0}(V_i - I_i Z_0) e^{\gamma z}$.

b) $V(z) = V_i \cosh \gamma z - I_i Z_0 \sinh \gamma z$,

$I(z) = I_i \cosh \gamma z - \frac{V_i}{Z_0} \sinh \gamma z$.

P.8-8 A d-c generator of voltage V_g and internal resistance R_g is connected to a lossy transmission line characterized by a resistance per unit length R and a conductance per unit length G.

a) Write the governing voltage and current transmission-line equations.

b) Find the general solutions for $V(z)$ and $I(z)$.

c) Specialize the solutions in part (b) to those for an infinite line.

d) Specialize the solutions in part (b) to those for a finite line of length ℓ that is terminated in a load resistance R_L.

P. 8-8 a) $-\frac{dV}{dz} = RI$, $-\frac{dI}{dz} = GV$.

$$\begin{cases} \frac{d^2V}{dz^2} = RGV, \\ \frac{d^2I}{dz^2} = RGI. \end{cases}$$

b) $V(z) = V_0^+ e^{-\alpha z} + V_0^- e^{\alpha z}$,

$I(z) = I_0^+ e^{-\alpha z} + I_0^- e^{\alpha z}$, $\alpha = \sqrt{RG}$.

$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} = R_0 = \sqrt{\frac{R}{G}}.$$

P. 8-8 a) $-\frac{dV}{dz} = RI$, $-\frac{dI}{dz} = GV$.

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$I(z) = I_0^+ e^{-\alpha z} + I_0^- e^{\alpha z}$, $\alpha = \sqrt{RG}$.

$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} = R_0 = \sqrt{\frac{R}{G}}.$$

P.8-9 A generator with an open-circuit voltage $v_g(t) = 10 \sin 8000\pi t$ (V) and internal impedance $Z_g = 40 + j30$ (Ω) is connected to a 50-(Ω) distortionless line. The line has a resistance of 0.5 (Ω/m), and its lossy dielectric medium has a loss tangent of 0.18%. The line is 50 (m) long and is terminated in a matched load. Find (a) the instantaneous expressions for the voltage and current at an arbitrary location on the line, (b) the instantaneous expressions for the voltage and current at the load, and (c) the average power transmitted to the load.

P. 8-9 Distortionless line: $R_0 = \sqrt{\frac{L}{C}} = 50\ (\Omega)$, $R = 0.5\ (\Omega/m)$,

$\tan\left(\frac{\sigma_d}{\omega\epsilon}\right) = \tan\left(\frac{G}{\omega C}\right) = 0.0018$.

$\longrightarrow \frac{G}{\omega C} \cong 0.0018$; $\frac{G}{C} = 8000\pi \times 0.0018 = 45.2 = \frac{R}{L}$.

$L = \frac{R}{G/C} = 0.011\ (H/m)$, $C = \frac{L}{R_0^2} = 4.42\ (\mu F/m)$.

$\alpha = \frac{R}{R_0} = 0.010\ (Np/m)$, $\beta = \omega\sqrt{LC} = 5.55\ (rad/m)$.

a) $V(z) = \frac{V_{g0} R_0}{R_0 + R_g} e^{-\alpha z} e^{-j\beta z} = \frac{50}{9+j3} e^{-0.01z} e^{-j5.55z}$; $I(z) = \frac{V(z)}{50}$.

$\therefore V(z,t) = 5.27\, e^{-0.01z} \sin(8000\pi t - 5.55z - 0.322)$ (V),

$I(z,t) = 0.105\, e^{-0.01z} \sin(8000\pi t - 5.55z - 0.322)$ (A).

b) At $z = 50$ (m): $V(50,t) = 3.20 \sin(8000\pi t - 0.432\pi)$ (V),

$I(50,t) = 0.064 \sin(8000\pi t - 0.432\pi)$ (A).

c) $(P_{av})_L = \frac{1}{2} Re\left|V_L I_L^*\right| = \frac{1}{2}(3.20 \times 0.064) = 0.102$ (W).

P.8-15 Consider a lossless transmission line.

a) Determine the line's characteristic resistance so that it will have a minimum possible standing-wave ratio for a load impedance $40 + j30$ (Ω).

b) Find this minimum standing-wave ratio and the corresponding voltage reflection coefficient.

c) Find the location of the voltage minimum nearest to the load.

P.8-15 For a lossless line, $Z_0 = R_0$.

$$|\Gamma|^2 = \left| \frac{(R_L - R_0) + jX_L}{(R_L + R_0) + jX_L} \right|^2 = \frac{(R_L - R_0)^2 + X_L^2}{(R_L + R_0)^2 + X_L^2}.$$

a) Set $\frac{\partial |\Gamma|^2}{\partial R_0} = 0. \longrightarrow R_0 = \sqrt{R_L^2 + X_L^2}$.

(A minimum S corresponds to a minimum $|\Gamma|$.)

For $Z_L = 40 + j30\ (\Omega)$, $R_0 = \sqrt{40^2 + 30^2} = 50\ (\Omega)$.

b) Min. $|\Gamma| = \sqrt{\frac{R_0 - R_L}{R_0 + R_L}} = \sqrt{\frac{50 - 40}{50 + 40}} = \frac{1}{3}$.

Min. $S = \frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} = 2$.

P.8-24 A dipole antenna having an input impedance of 73 (Ω) is fed by a 200-(MHz) source through a 300-(Ω) two-wire transmission line. Design a quarter-wave two-wire air line with a 2-(cm) spacing to match the antenna to the 300-(Ω) line.

P.8-24 $f = 2 \times 10^8$ (Hz),

$\lambda = 1.5$ (m). $\longrightarrow \ell = \frac{\lambda}{4} = 0.375$ (m).

Characteristic impedance of quarter-wire two-wire transmission line, $Z_0 = \sqrt{73 \times 300} = 148\ (\Omega)$.

For a lossless air line, from Eqs. (8-23) and (8-24),

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{\pi}\sqrt{\frac{\mu_0}{\epsilon_0}} \cosh^{-1}\left(\frac{D}{2a}\right).$$

$$148 = 120 \cosh^{-1}\left(\frac{D}{2a}\right).$$

Given $D = 2$ (cm) $\longrightarrow$ a (wire radius) $= 0.54$ (cm).

P.8-28 A load impedance can be matched to a transmission line also by using a single stub placed in series with the load at an appropriate location, as shown in Fig. 8-14. Assuming that $Z_L = 25 + j25$ (Ω), $R_0 = 50$ (Ω), and $R_0' = 35$ (Ω), find d and ℓ required for matching.

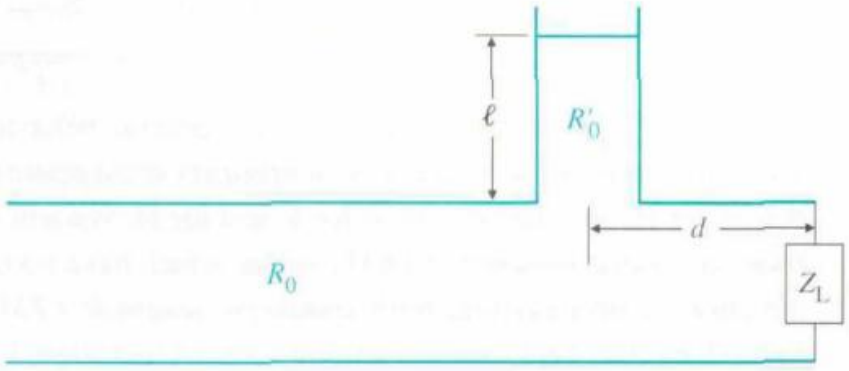


FIGURE 8-14 Impedance matching by a series stub (Problem P.8-28).

P.8-28 Use Smith chart as an impedance chart. —— Same construction as that in problem P.8-25, except point P_{sc} would be at the extreme left (marked by a ×) and the $g=1$ circle becomes a $r=1$ circle.

P_1: $z_L = (25 + j25)/50 = 0.5 + j0.5$.

Two possible solutions:

At P_3: $z_{s3} = 1 + j1$.
$\longrightarrow d_3 = (0.162\lambda - 0.088\lambda) = 0.074\lambda$.

At P_2: $z_{s3} = 1 - j1$.
$\longrightarrow d_2 = (0.338\lambda - 0.088\lambda) = 0.250\lambda$.

To achieve a match with a series stub having $R_0' = \frac{35}{50} R_0$, we need a normalized stub reactance $-j\frac{50}{35} = -j1.43$ for solution corresponding to P_3. From Smith chart we find the required stub length $\ell_3 = 0.347\lambda$.

Similarly for solution corresponding to P_2, a stub length with a normalized reactance $+j1.43$ is needed, which requires a stub length $\ell_2 = 0.153\lambda$.